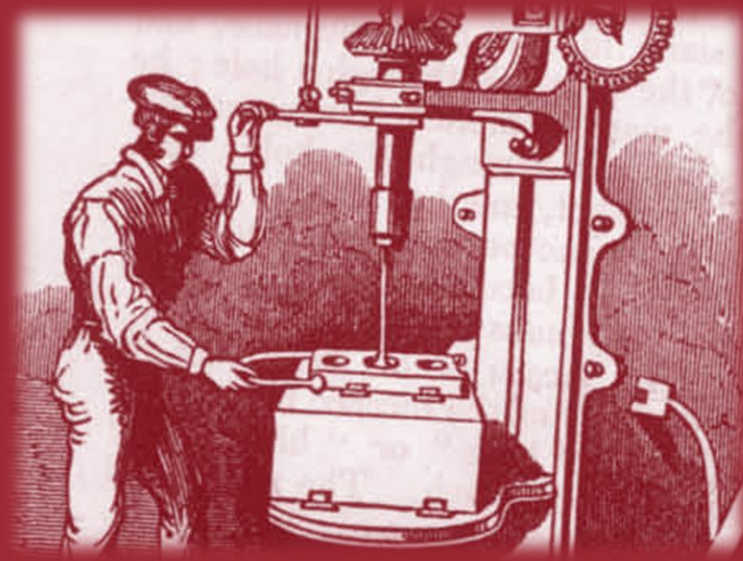




316-221 Metallurgy

Chapter 2: Metal & Alloys



Contents

∞ Foundation of Metallurgy

∞ Type of Carbon Steels

∞ Type of Stainless steels

∞ Type of Cast Irons

∞ Nonferrous Metals

Foundation of Metallurgy

∞ The metal used in industry is divided 2 groups

- **Ferrous metals**

- metal with base metals and other elements are mixed in different types of steel

- **Non-ferrous metals**

- metal with other elements at the non-ferrous base metals

Ferrous metals

- ❑ **Steel production process**
- ❑ **Classification of steel**
- ❑ **Steel used in industry**



Steel Production Process

□ Steel making process

- Reduction of iron compounds to molten iron
- Refining
- Casting
- Rolling
- Heat treatment



Steel Production Process

□ Iron ores



1cm

Magnetite

Magnetite (Fe_3O_4) have iron about 70%



1cm

Hematite

Hematite (Fe_2O_3) have iron about 70%



1cm

Pyrite

Pyrite (FeS_2) have **iron** about 45%

Steel Production Process



Siderite

**Siderite (FeCO_3) have
iron about 60%**

Source: <http://www.musee.ensmp.fr/mineral/11087x.jpg>

Source: <http://www.gc.maricopa.edu/earthsci/imagearchive/limonite.htm>



Limonite

**Limonite (HFeO_2) have
iron about 60%**

Steel Production Process

- Reduction of iron compounds to molten iron
 - Changing iron ores to high purity iron ore (% steel increase) by removing impurities from iron ore (mineral dressing)



Coarse Crusher



Fine Crusher



Ball Mill



Spiral Classifier



Magnetic Separator



Flotation Machine(20 cells)



Spiral Separator(3PCS)



Qualified concentrate containing

Steel Production Process

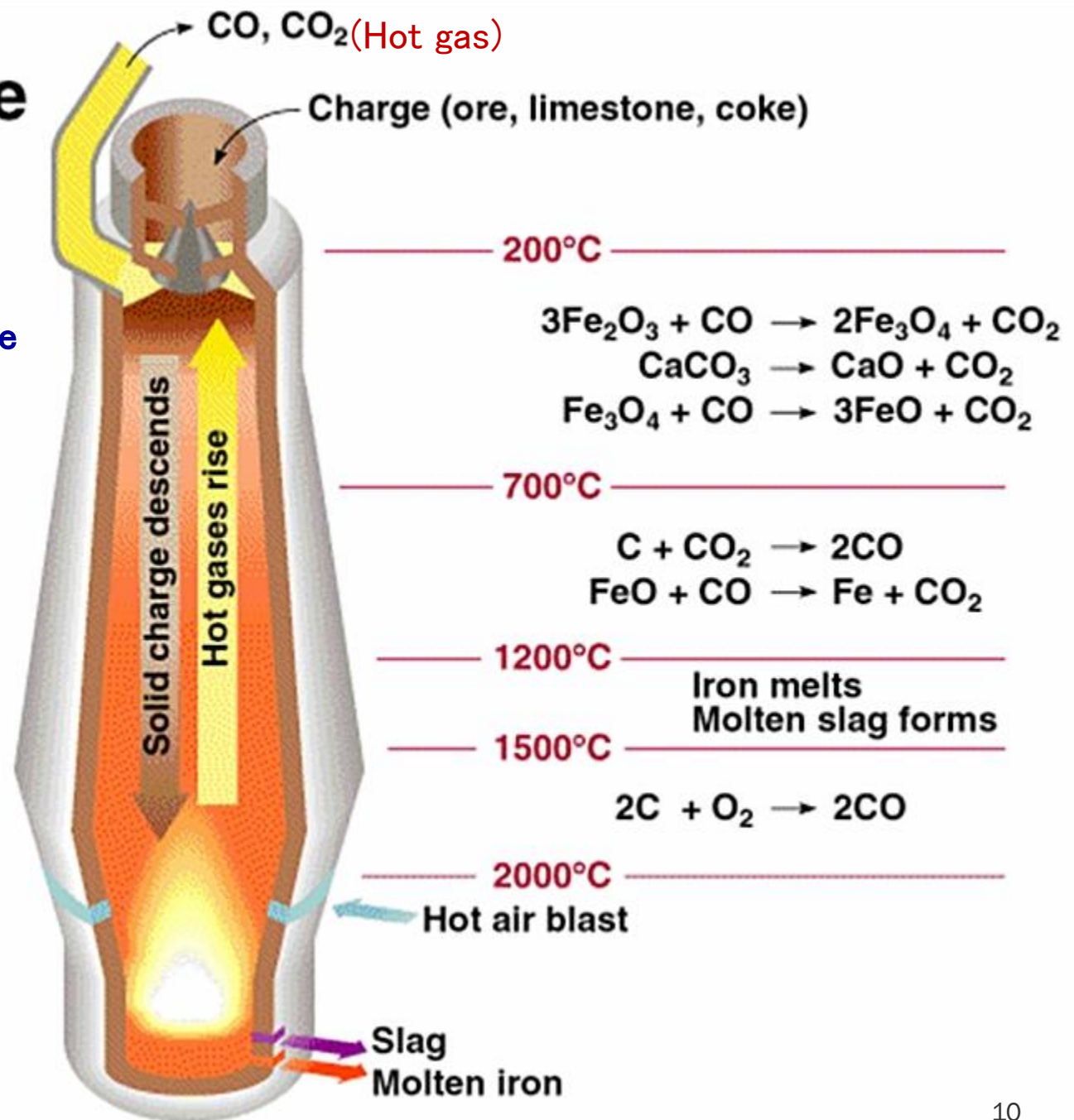
□ Process Steel making

- Make in the furnace at high temperature is call “**Blast furnace**”
- The fuel used in this process is **coal**
- Use flux is **limestone** (CaCO_3) in cover surface of iron melt and gas displace

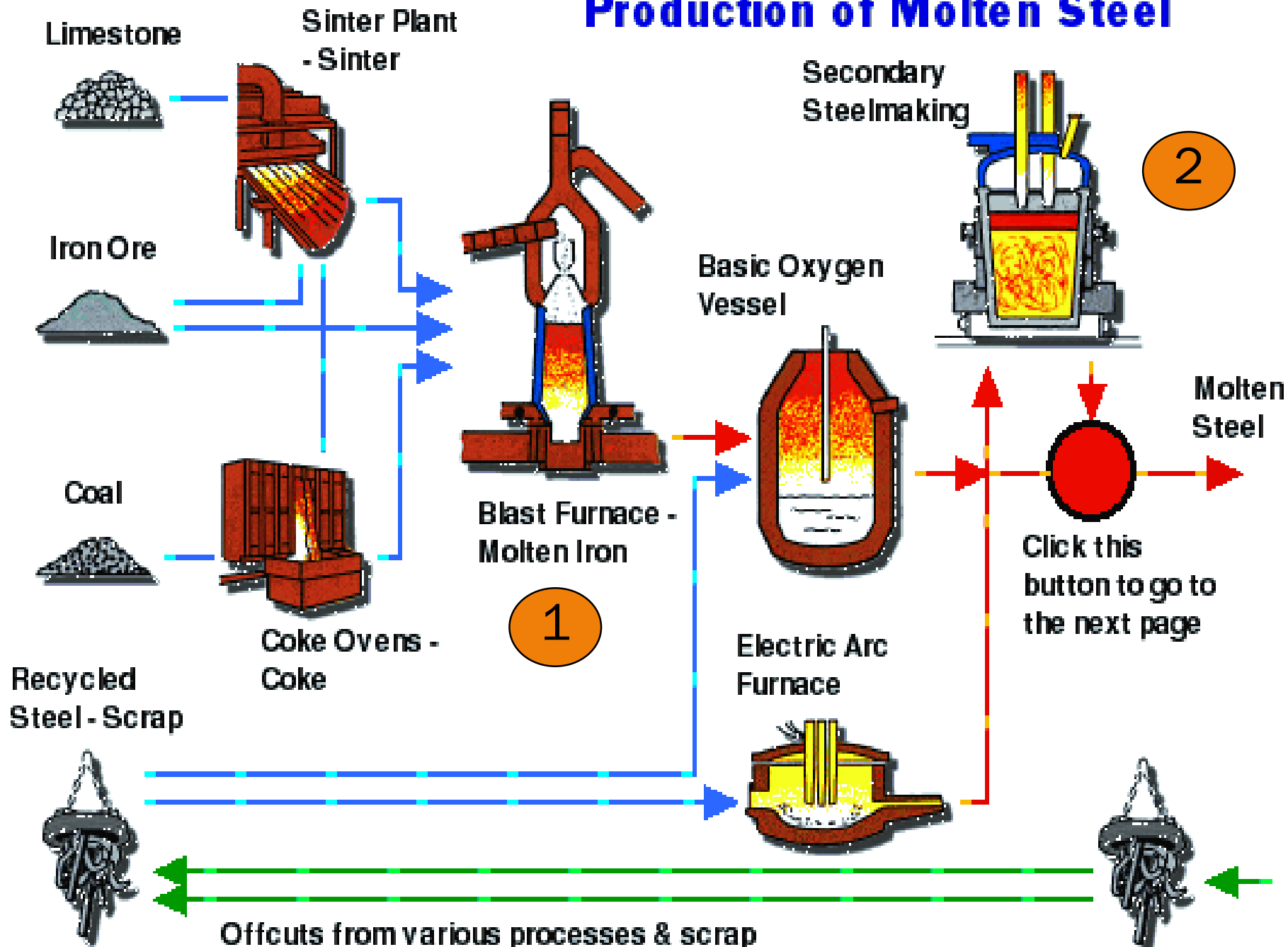
Blast Furnace

Pig iron melt will compose of element follow to:

- Carbon ~ 3.5%
- Silicon ~ 2.0%
- Manganese ~ 1.5%
- Sulphurs ~ 1.0%
- Phosphorus ~ 1.0%



Production of Molten Steel



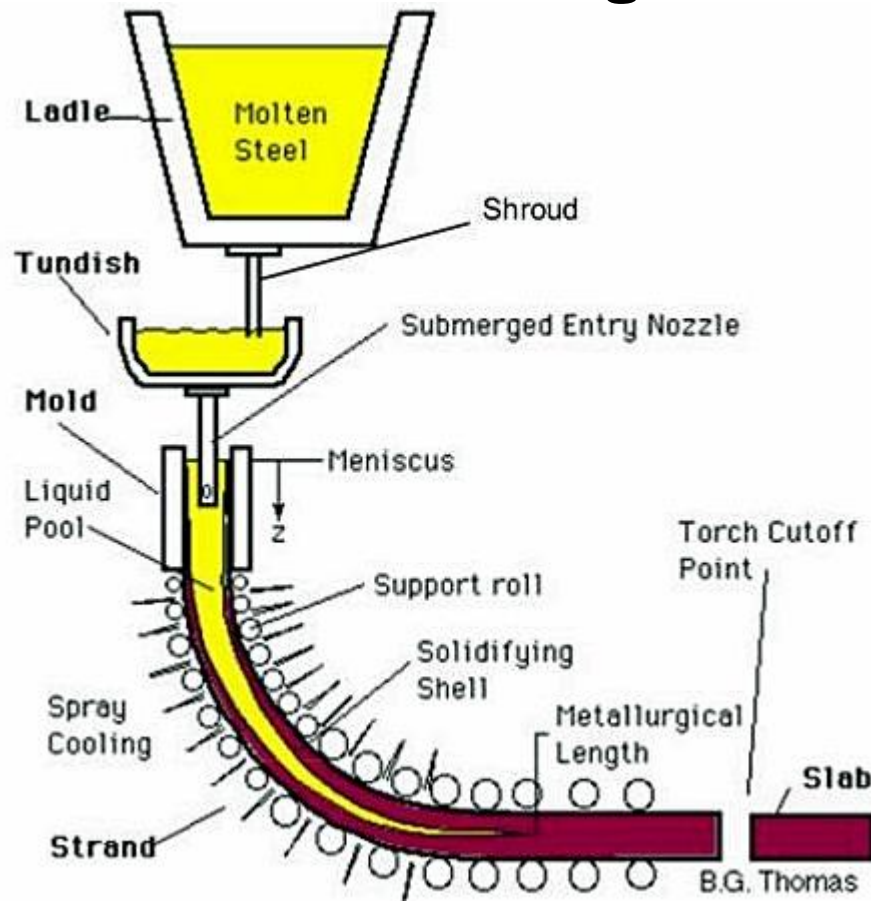
Casting

- ❑ The casting of iron by pouring metal melt into a mold and solidification, **before pouring** , it will improve composition again
 - Continuous casting
 - Ingot casting



Casting

Continuous casting



- Pour the liquid metal to flow through mold continuous
- Semi-finished products such as **Billet, Bloom, Slab**

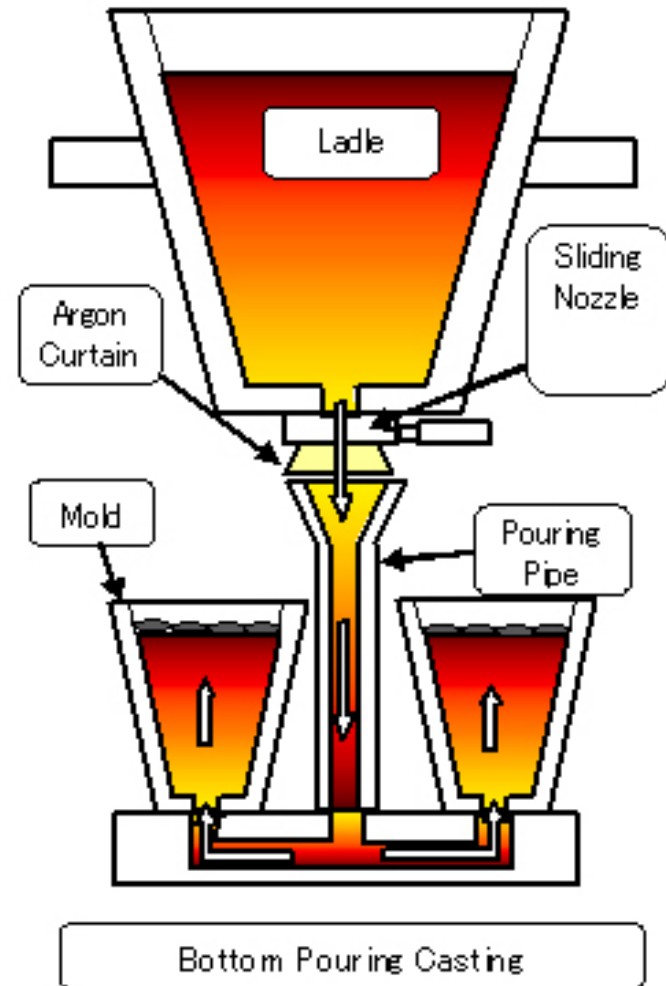
Casting



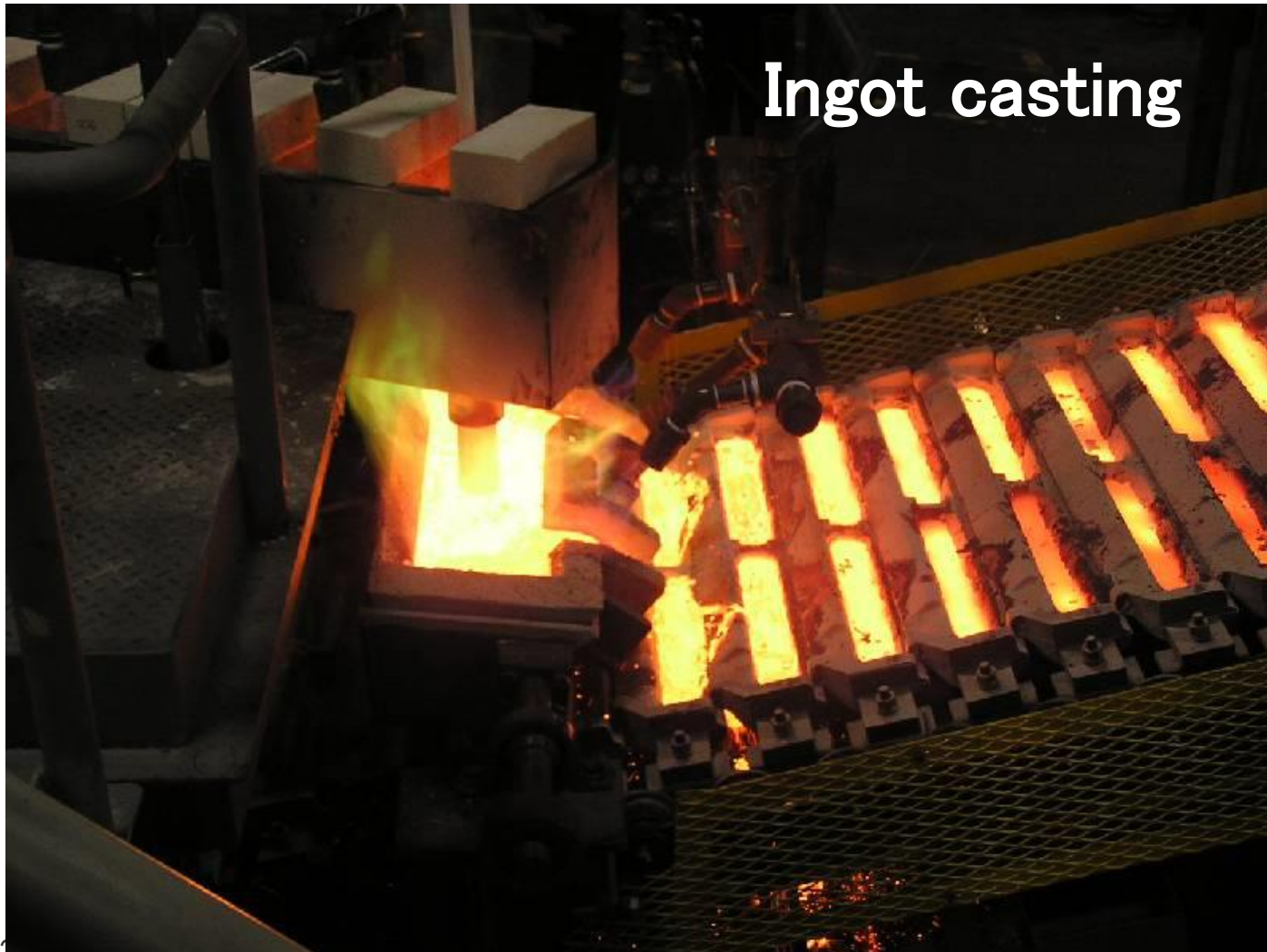
Casting

❑ Ingot casting

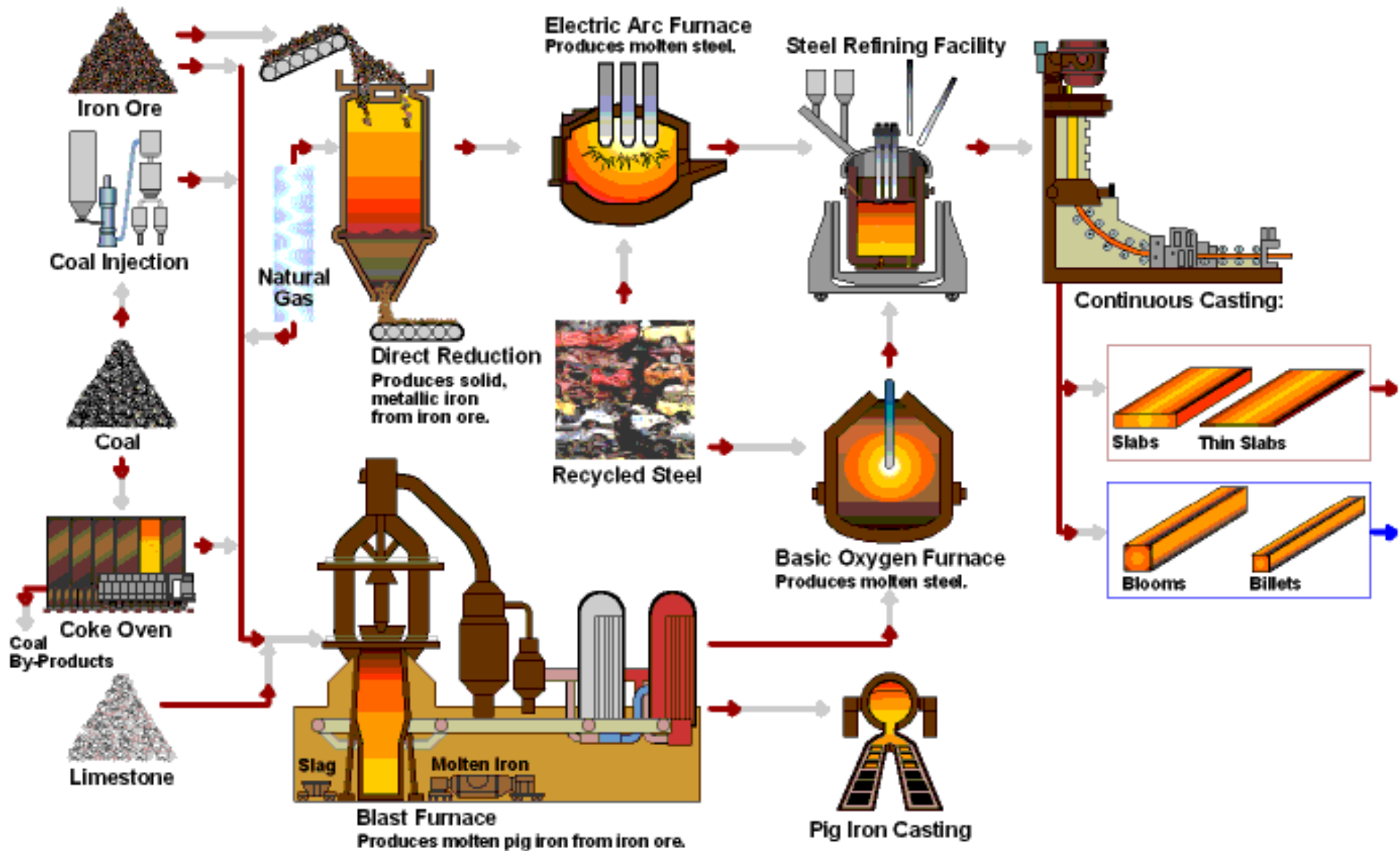
- The casting by pouring metal into mould stationary
- Products rods be processed into products for various applications



Casting



Ingot casting



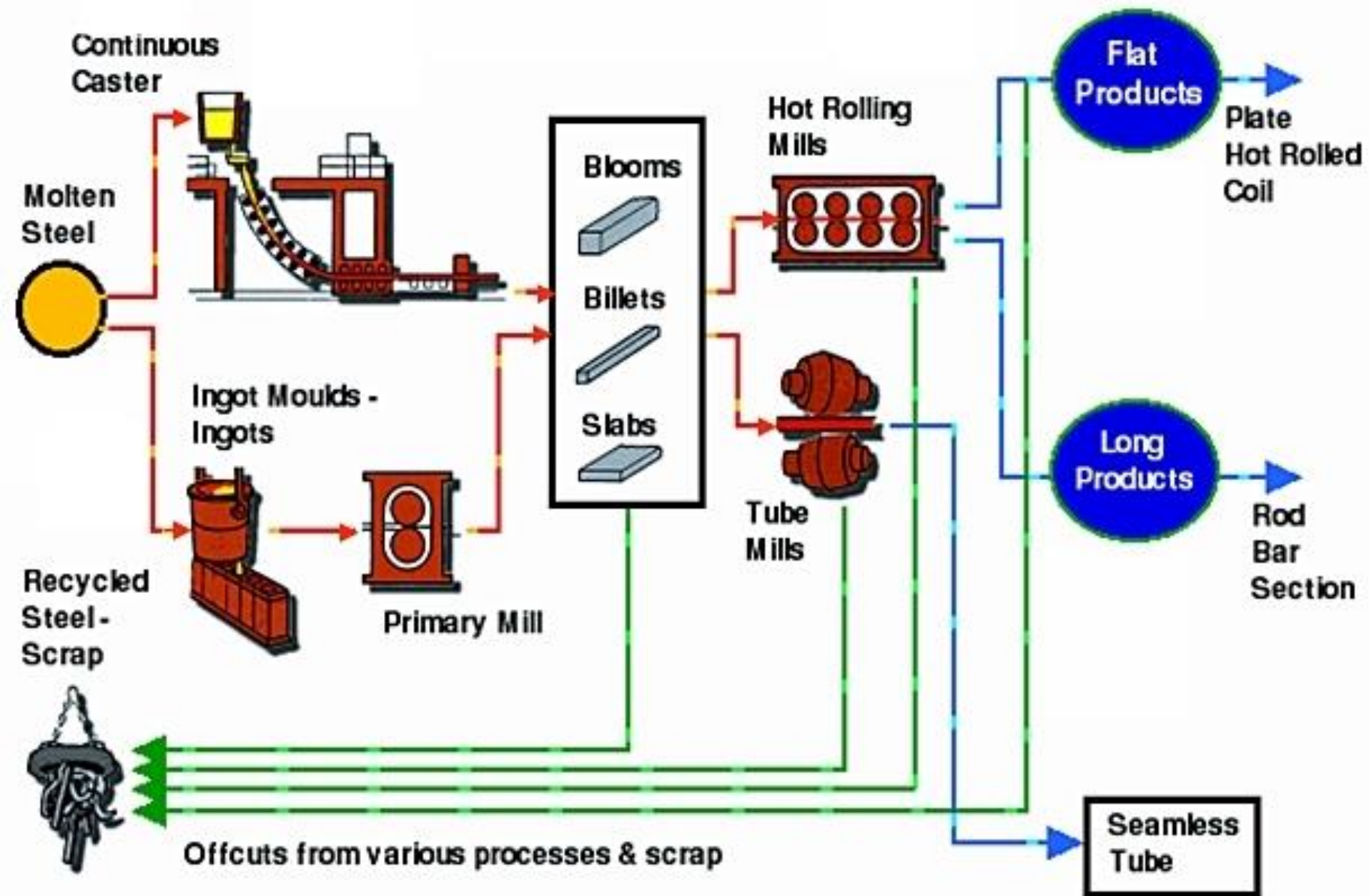
Steel Production Process

Rolling and Mechanical treatment

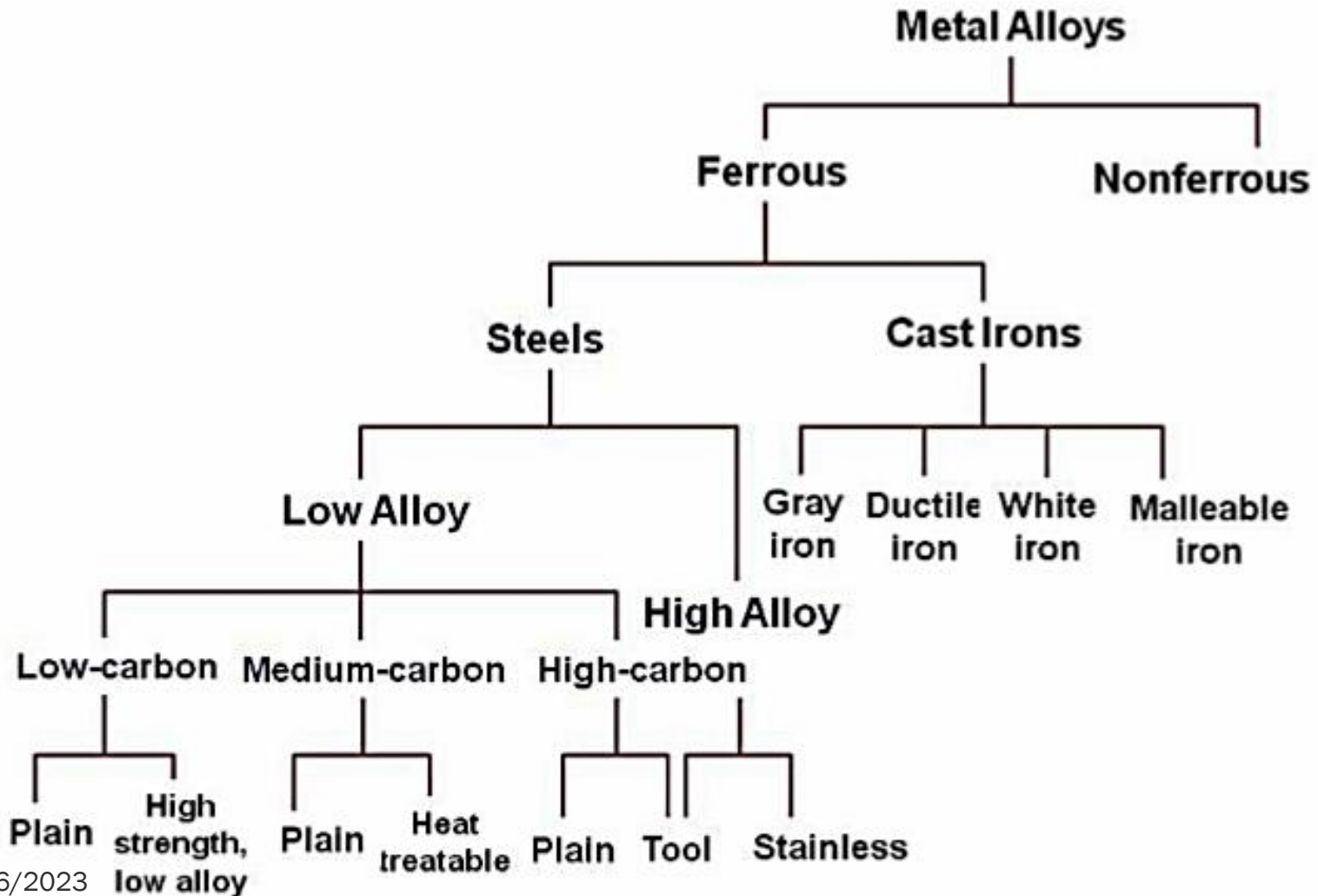
- Hot rolling
- Cold rolling
- Forging
- Extruding

Heat treatment

- Improved mechanical properties by heat treatment process



Classification of Ferrous Alloys



Classification of Ferrous Alloys

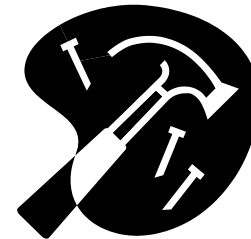
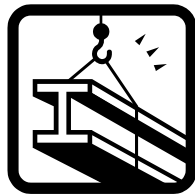
□ Divided by the amount of carbon is 2 groups

- **Steels**

- carbon lower 2 % ($C < 2\text{wt.}\%$)

- **Cast iron**

- carbon higher or equal 2 % ($C \geq 2\text{wt.}\%$)



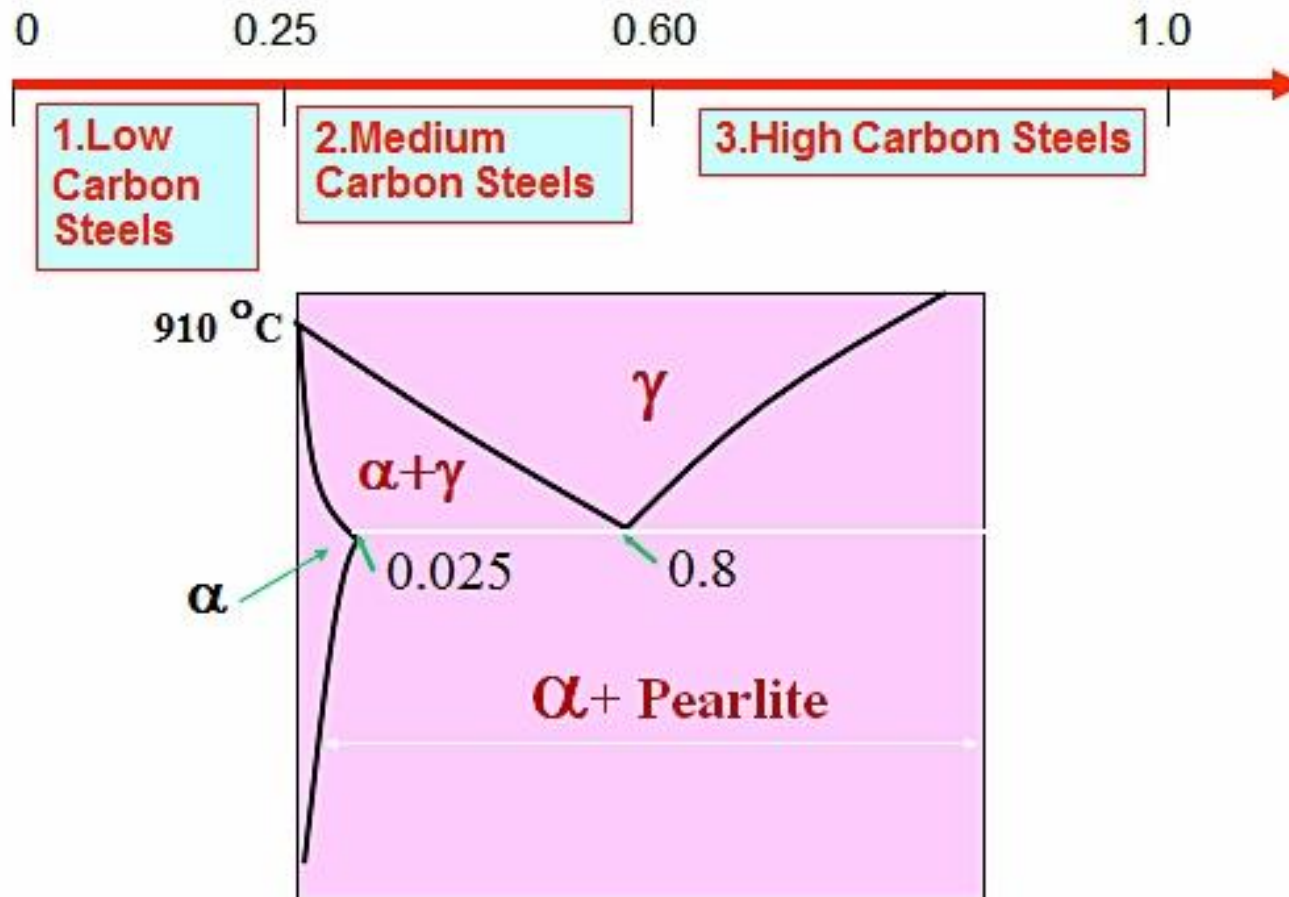
Classification of Ferrous Alloys

□ Steels

- steels are contained of carbon lower 2 wt.%, e.g. carbon steel, alloy steel etc.
- main composition of steels are Carbon (C) and Manganese (Mn) and Phosphorus (P), Silicon (Si), Sulphur (S) are little.
- widely used in construction and other applications because of their high tensile strengths and low costs.

Classification of Ferrous Alloys

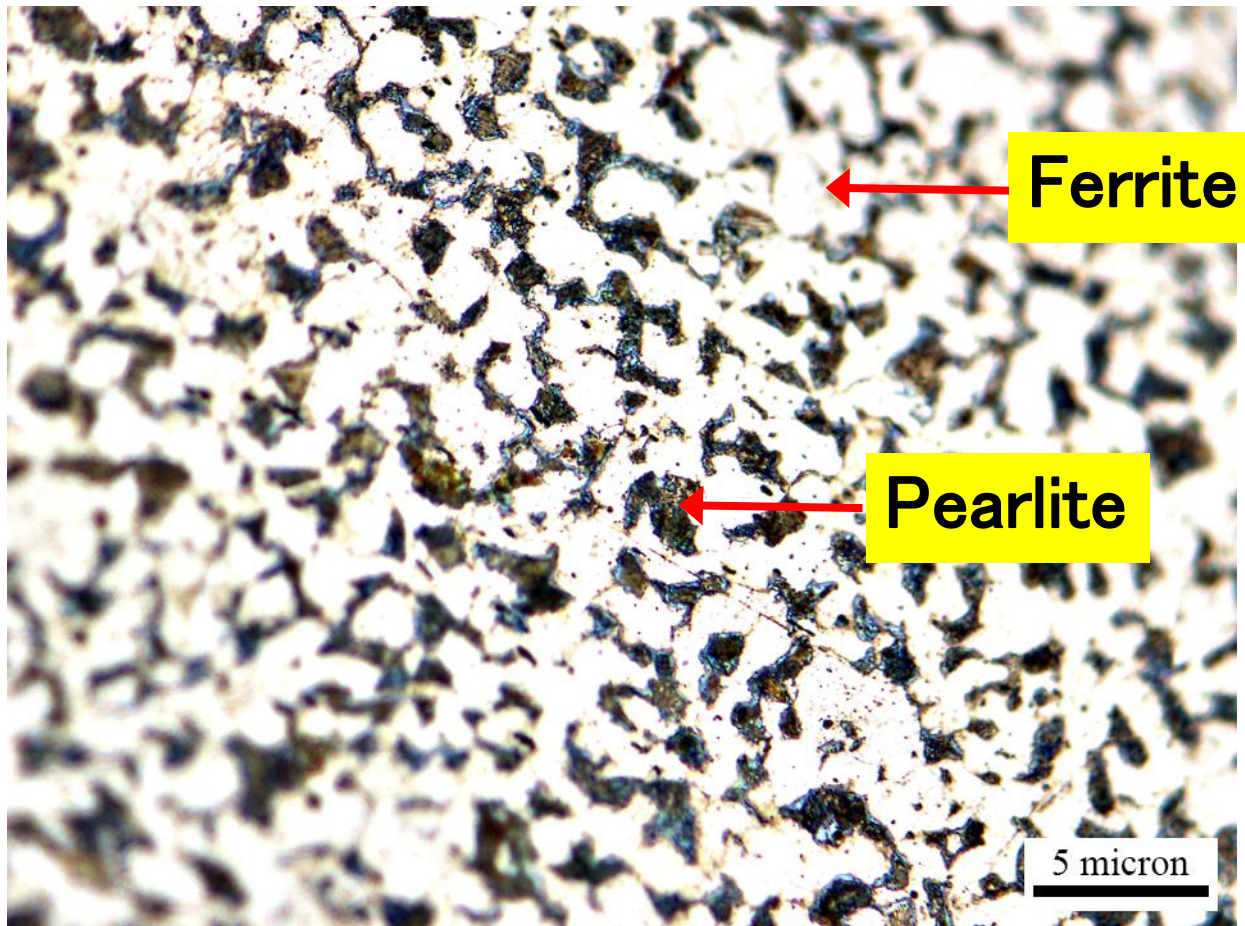
□ Carbon steels >> 3 types



Low Carbon Steels

- ❑ Steels are contained of carbon lower **0.25 wt.%**
- ❑ **Example:** plain carbon steel, high strength low alloy (HSLA) with other elements such as **Cu, V, Ni, Mo**
- ❑ High toughness and low strength
- ❑ **Applications:** sheet iron, channels, iron wire for building

Low Carbon Steels

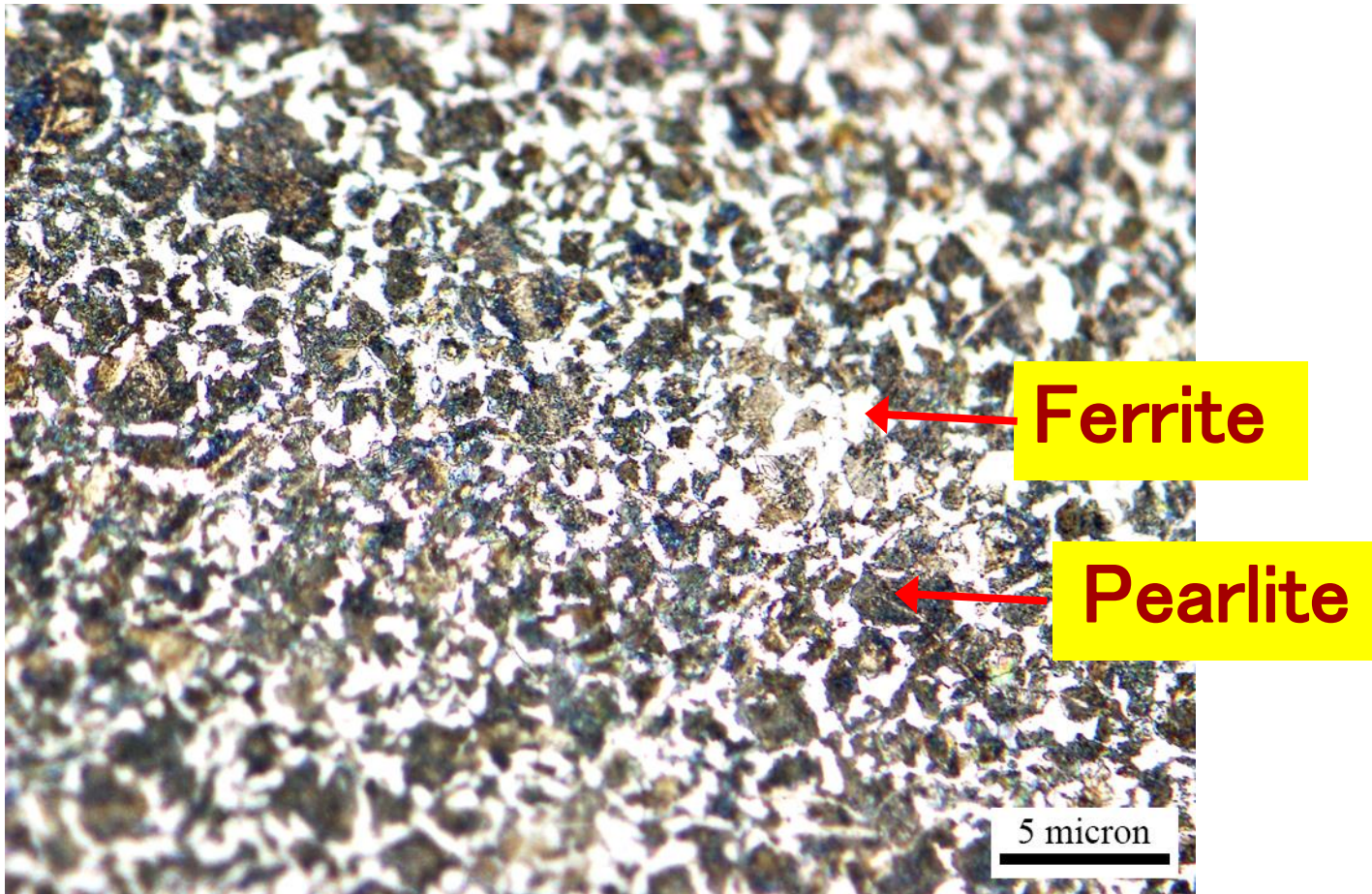


Low Carbon steel 0.2wt.%C

Medium Carbon Steels

- ❑ Steels are contained of carbon between **0.25–0.6 wt.%**
- ❑ Strength higher than low carbon steels **but** lower toughness
- ❑ **Applications:** wheels , screw, gear, die

Medium Carbon Steels

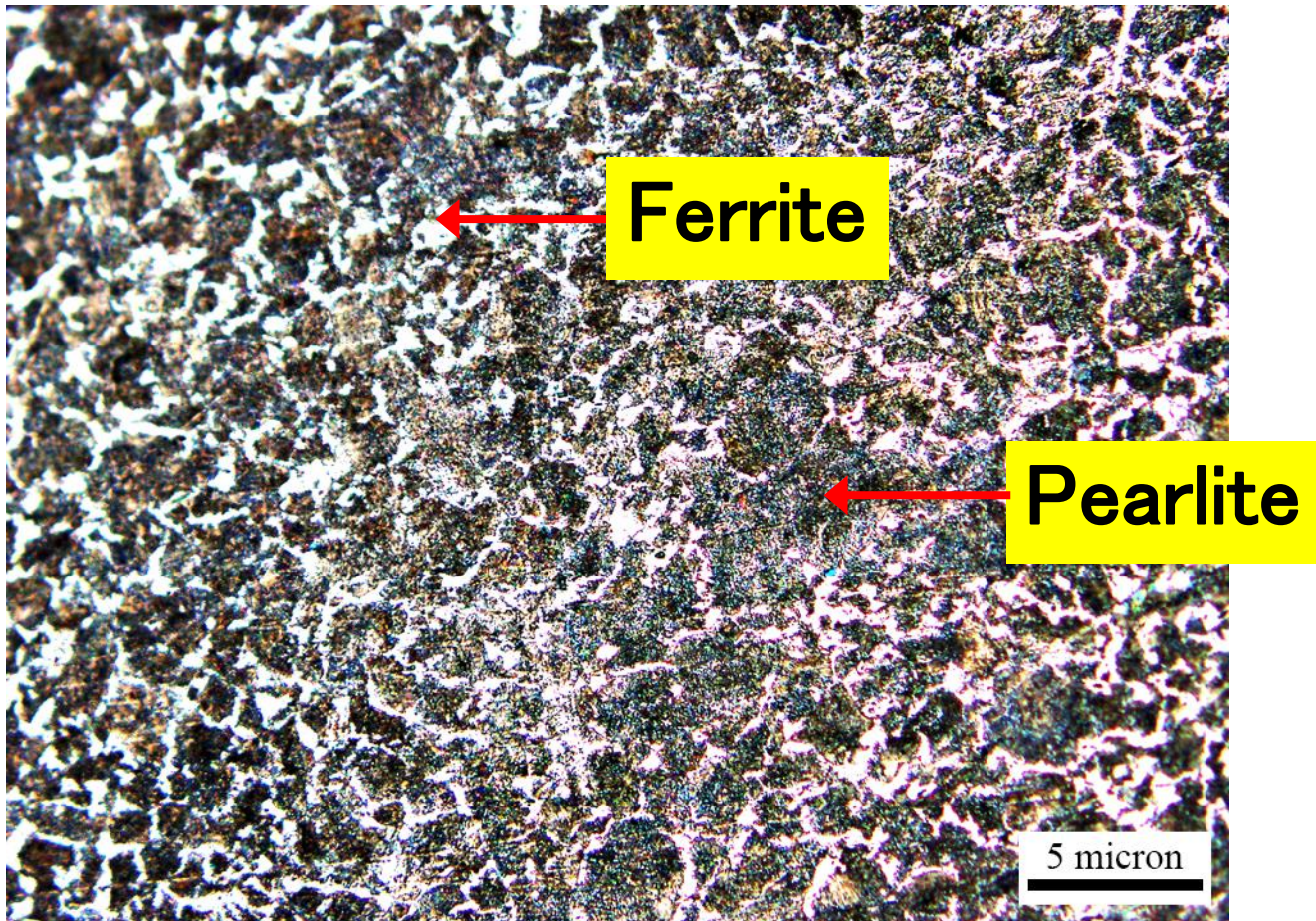


Carbon steel 0.6 wt.% C

High Carbon Steels

- ❑ Steels are contained of carbon between **0.6–1.0 wt.%**
- ❑ Higher strength **but** lower toughness (brittle)
- ❑ Can be hardened as well
- ❑ **Applications:** knife, ball bearing, springs, tool bits

High Carbon Steels



High Strength Low Alloy Steels

- ❑ **High Strength Low Alloy Steels** are contained of carbon below **0.3 wt.%**
- ❑ Can increase strength, corrosion resistance, toughness by adding elements for creating carbide compounds
- ❑ Can be forming as well
- ❑ An easy oxidation at high temperatures

High Alloy Steels

□ Stainless Steels

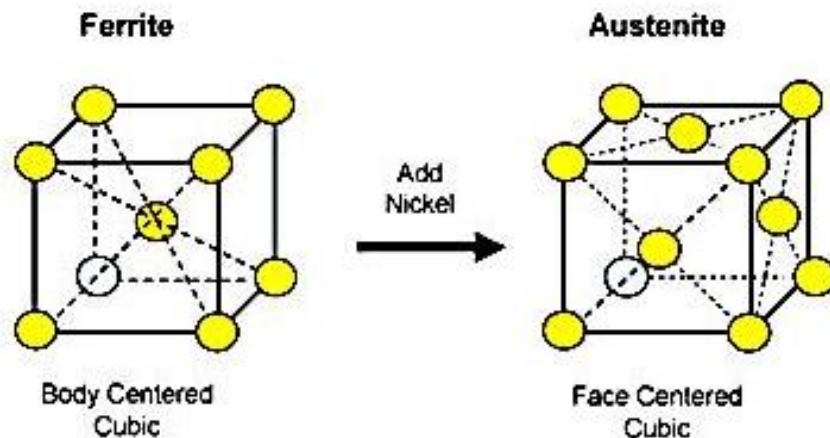
- Steels are contained of **chromium (Cr)** higher 11 wt.% (Cr > 11wt.%) with other elements such as **Ni , Mn, C, Mo**
- **Cr** combines with **oxygen** in the air, cause occur "**chromium oxide**" this is a thin film coating on the surface of the steel, to qualify for corrosion resistance

Classification of Stainless Steels

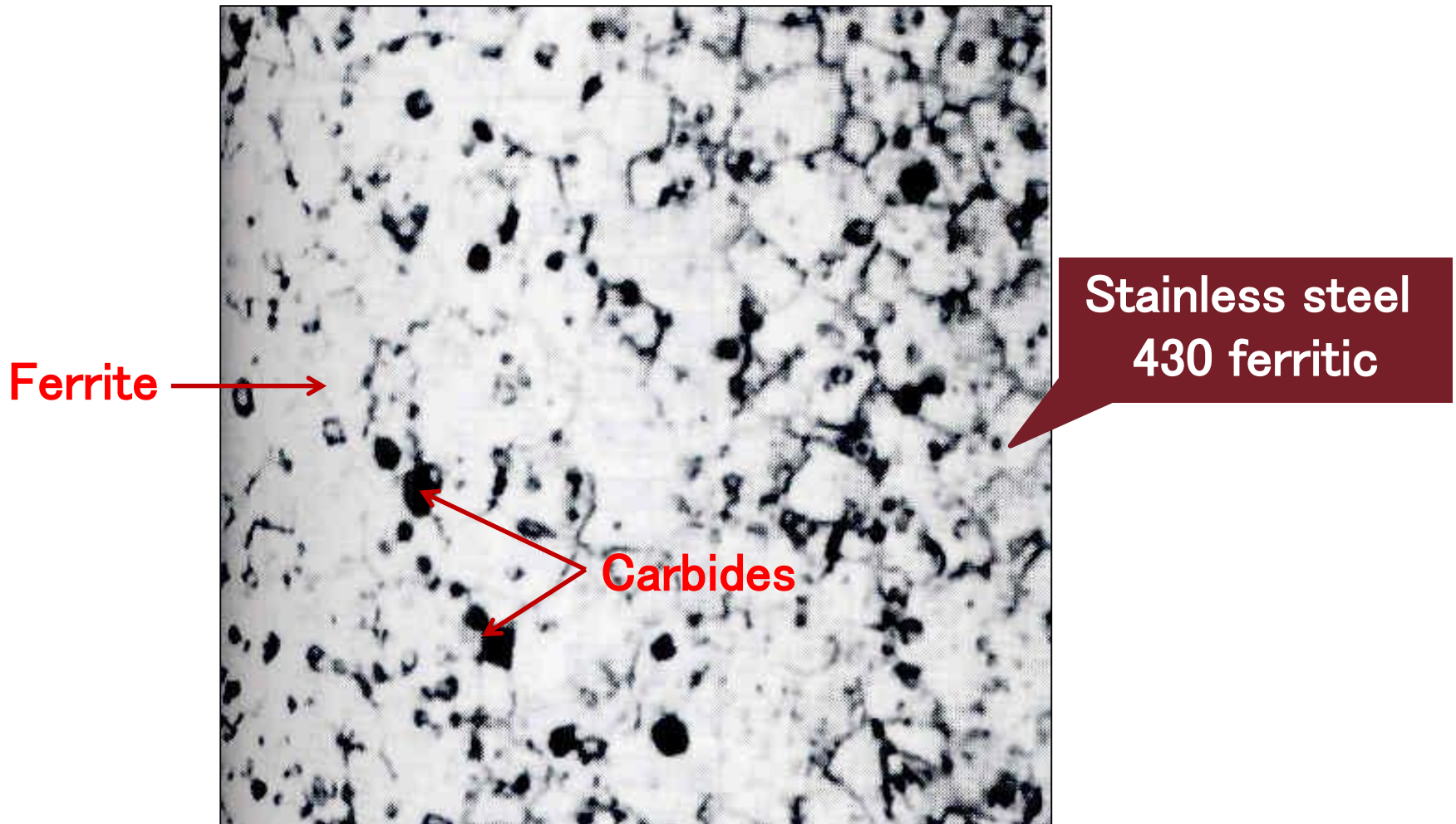
- ☐ Ferritic
- ☐ Austenitic
- ☐ Martensitic
- ☐ Precipitation Hardening
- ☐ Duplex

Ferritic Stainless Steels

- ❑ **Cr** elements is about 12–17wt.%
- ❑ Ferrite crystal structure (**BCC**) with magnetic properties
- ❑ Medium **yield strength** and **tensile strength**
- ❑ High **elongation**



Ferritic Stainless Steels



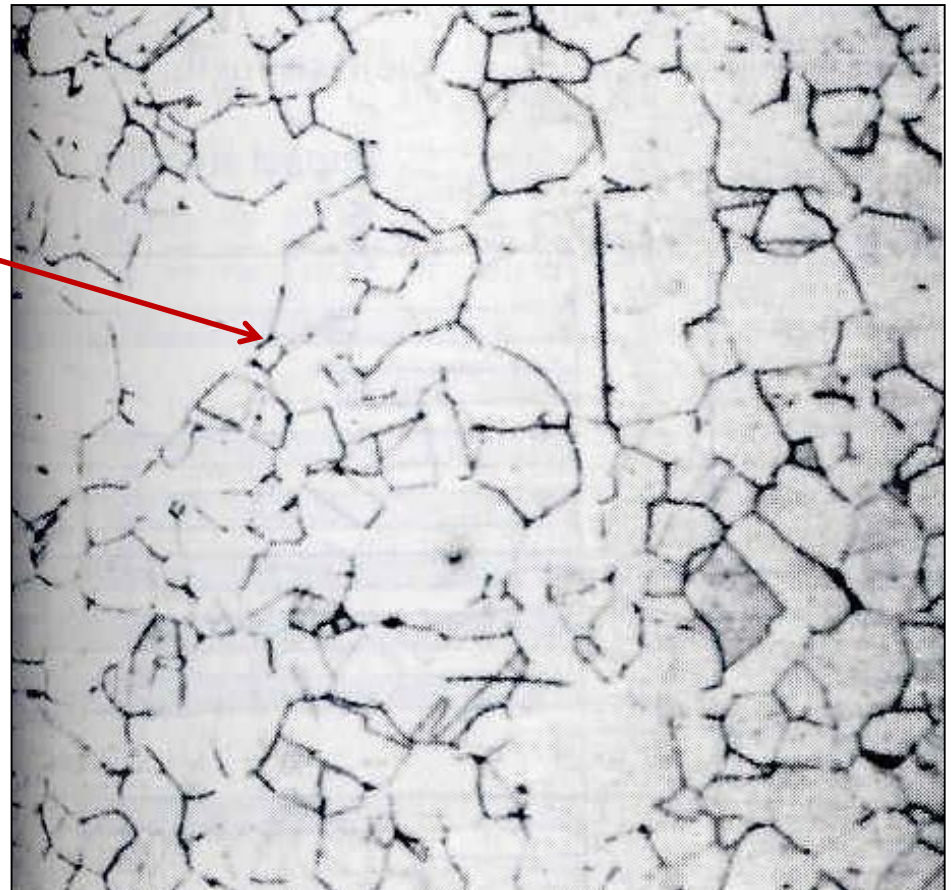
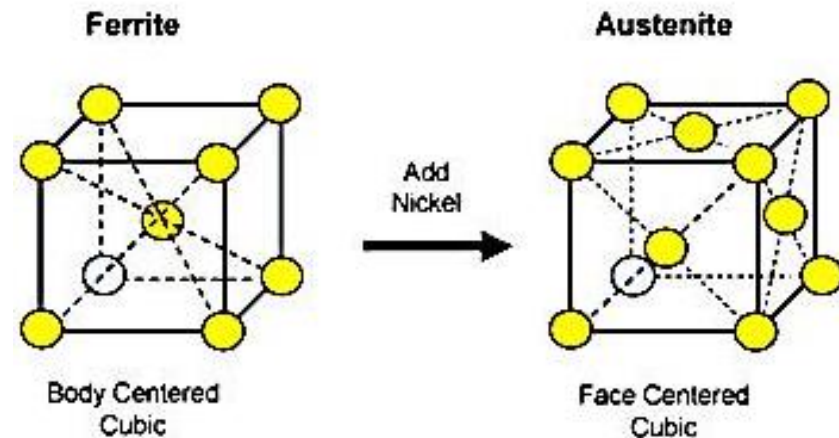
Austenitic Stainless Steels

- ❑ **Cr** elements is about 12–17wt.% and 6–22 **Ni** wt.%
- ❑ Austenitic crystal structure (**FCC**) with not magnetic properties
- ❑ **Ni** increase corrosion resistance and oxidation resistance at high temperature
- ❑ **Yield strength** close to ferritic group but **tensile strength** and **elongation is higher**
- ❑ 304 stainless steel very popular: **pot, spoon, tray**

Austenitic Stainless Steels

316 Stainless steel: Austenitic

Austenite grains



Martensitic Stainless Steels

- ❑ **Cr** elements is about 11.5–18wt.% and 0.1–1.0 **C** wt.%
- ❑ Martensite crystal structure **and** can hardening process
- ❑ High hardness
- ❑ **Corrosion resistance less than** austenitic and ferritic

Martensitic Stainless Steels



440 Stainless steel: Martensitic

carbide

**Stainless steel 15-5PH
low carbon martensitic**



Precipitation Hardening Stainless Steels

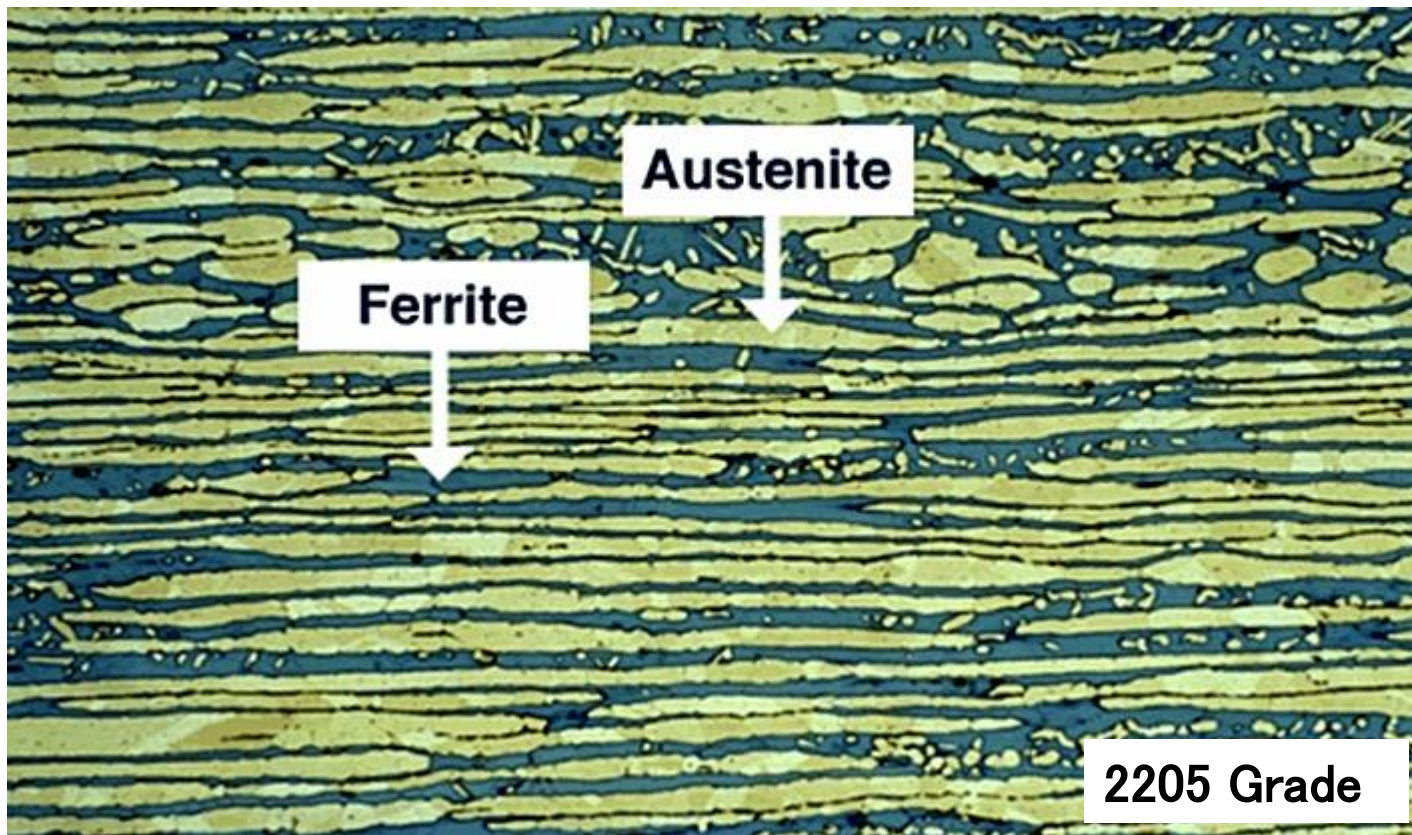
- ❑ **Cr** elements is about 15–18wt.%, 3–8 **Ni** wt.% and **Mo** (little)
- ❑ High strength and high cost
- ❑ High **Corrosion and temperature resistance**
- ❑ Austenite or martensite matrix and precipitation hardening are occurred composition of **Cu, Al, Ti**
- ❑ **Example:** PH13–9Mo, AM–350: **pump, valves**

Duplex Stainless Steel

- ❑ High **Cr** elements is about 18–28wt.%, 4.5–8 **Ni** wt.% and 2.5–4 **Mo** wt.%
- ❑ **Ferritic** and **Austenitic** crystal structure
- ❑ **Very good corrosion resistance**
- ❑ Can welding and forming
- ❑ **Example:** 2205Grade : Heat Exchangers, Chemical Tanks, Refinery

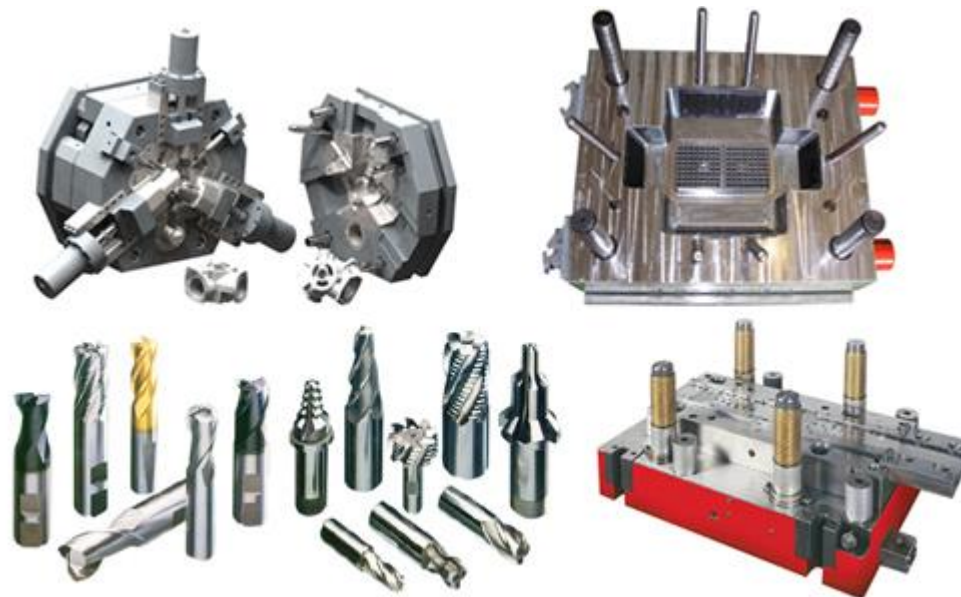
Duplex Stainless Steel

Microstructure of roughly 50% austenite and 50% ferrite



Tool Steels

- ❑ **Tool steels** used for forming tools e.g. drill, mould, milling cutters
- ❑ Element composition: **C , W , Mo , Cr , Co , Ni , Mn , Si , Ti , V**



Tool Steels: Properties

- ❑ Good wear and friction
- ❑ Sharp (milling cutters)
- ❑ Good heat resistance
- ❑ A toughness and stability, even when heated
- ❑ Easily forming
- ❑ Can be hardende: high Hardenability

Types of Tool Steels

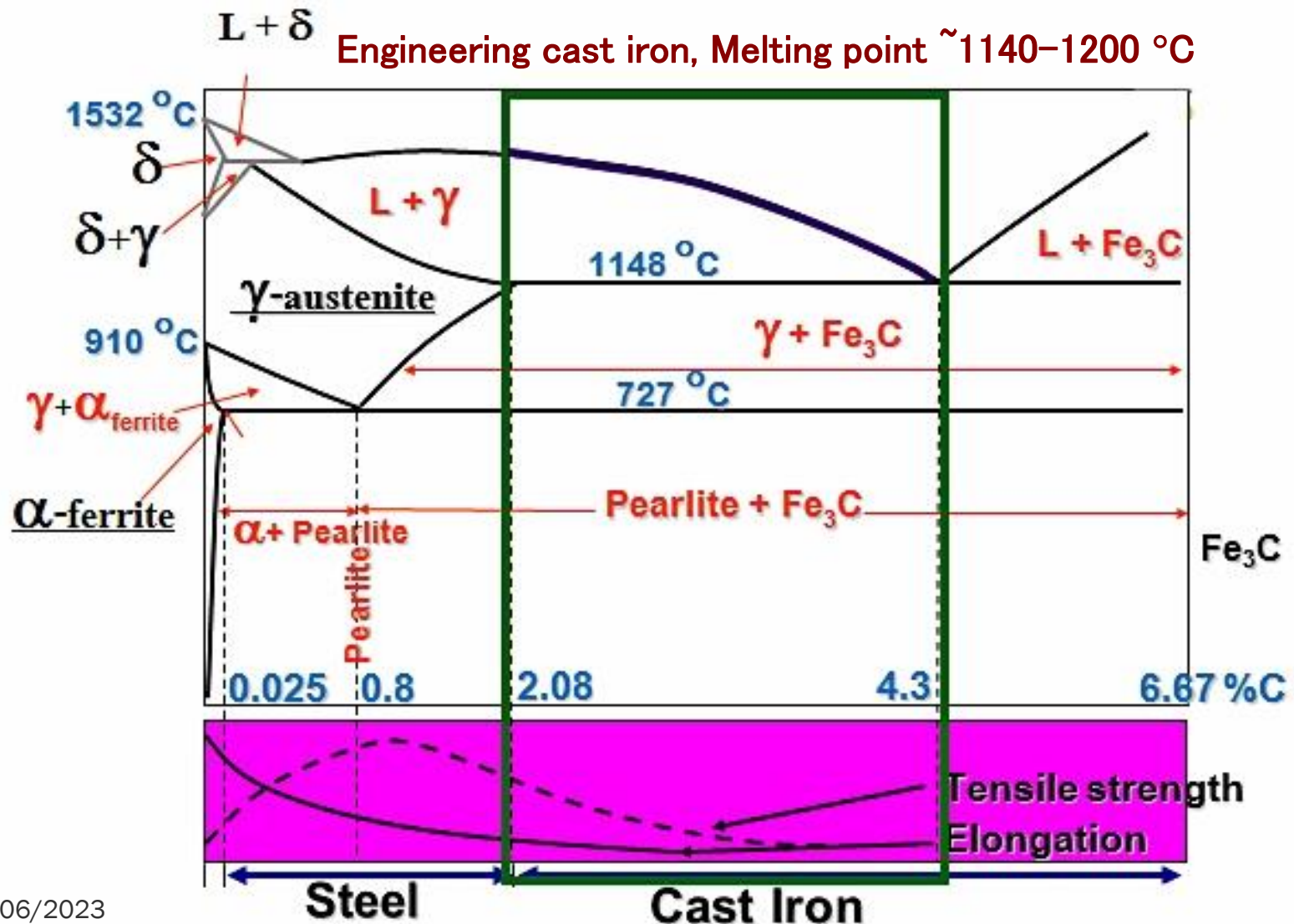
- ❑ High speed steel, HSS
- ❑ Hot work tool steel: **Hot forming**, (main elements: Cr, W, Mo)
- ❑ Cool work tool steel
- ❑ Impactance tool steel
- ❑ High carbon tool steel
- ❑ Plastic molding tool steel (low alloys)

Cast Irons

- ❑ Cast iron is a group of iron–carbon alloys with a carbon (C) content greater than 2% and silicon (Si) content little
- ❑ Higher carbon content in cast irons, the melting point lower pure irons and steels, therefore this cast irons are suitable for using in casting work.



Cast Irons



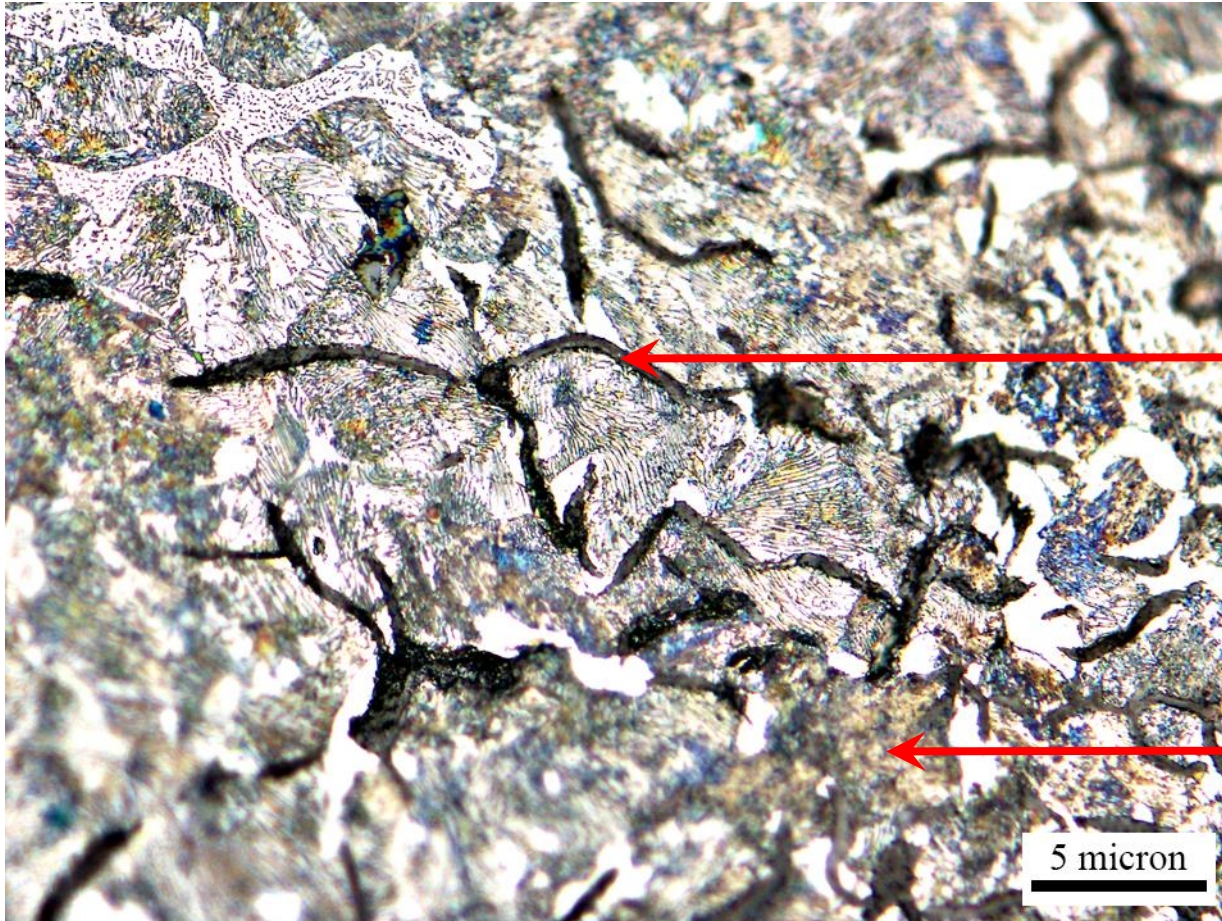
Type of Cast Irons

- ☐ Gray cast iron
- ☐ White cast iron
- ☐ Ductile cast iron
- ☐ Malleable cast iron
- ☐ Alloys or Special cast iron

Gray Cast Irons

- ❑ Gray cast iron is similar composition with pig iron at smelting form blast furnace by carbon content about 2–4 wt.%
- ❑ Occurring form slow cooling rate, carbon can form in Graphite Flake
- ❑ Low melting point and cost, good compressive force resistance
- ❑ Low strength about 200–250 MPa
- ❑ Low ductility

Gray Cast Irons



Gray Cast Iron structure
standard ASTM type A

Graphite flake

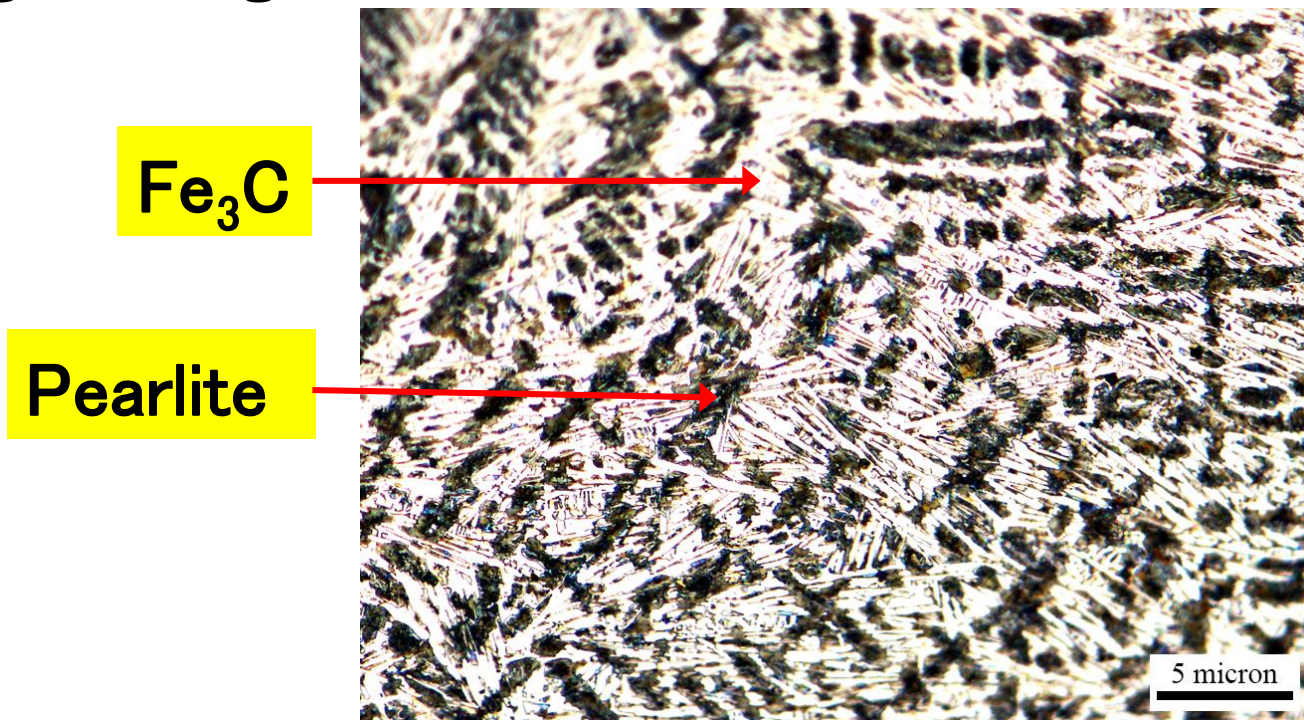
Pearlite

White Cast Irons

- ❑ Cast iron with **carbon** content about 1.7–3.0wt.%, 1%Si, 0.5%Mn and $P < 0.1\%$, $S < 0.2\%$
- ❑ Occurring from rapid cooling rate, carbon form in **solid solution**, therefore **can't form graphite**
- ❑ Carbon (**C**) mixed with steel (**Fe**) occur **Fe_3C** with white colour, therefore it call White Cast Irons

White Cast Irons

- ❑ Microstructure at the room temperature with consist of **Pearlite** and **Cementite (Fe_3C)**, It's call **Ledebulite**
- ❑ High strength but brittle

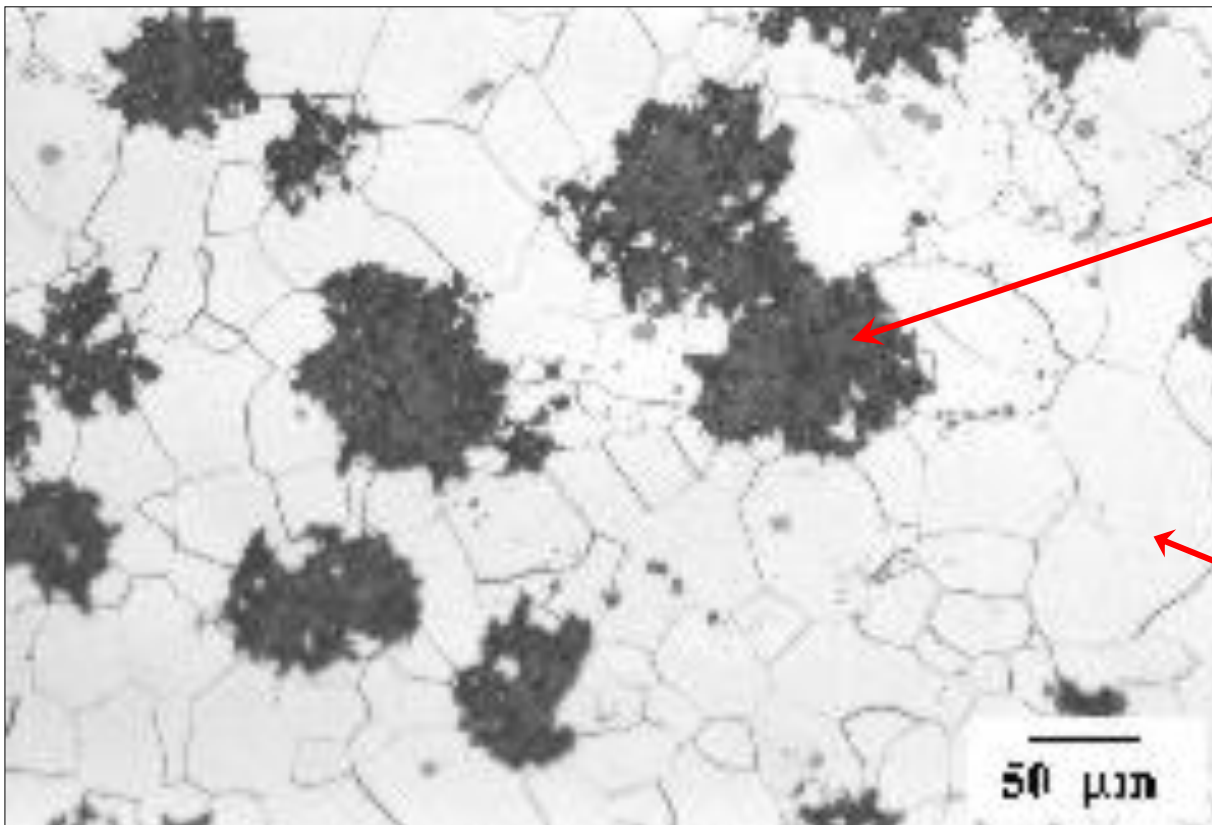


Malleable Cast Iron

- ❑ Received from white cast iron to bake at **1600 F** for **24–48 hr**, resulting in the disintegration of **iron carbide** and **carbon atom** joint and form to round small graphite
- ❑ Structure may be consist of **Ferrite** or **Pearlite**, result malleable cast iron shows good ductility and strength

Malleable Cast Iron

Malleable cast iron structure shows rosette graphite in ferrite matrix



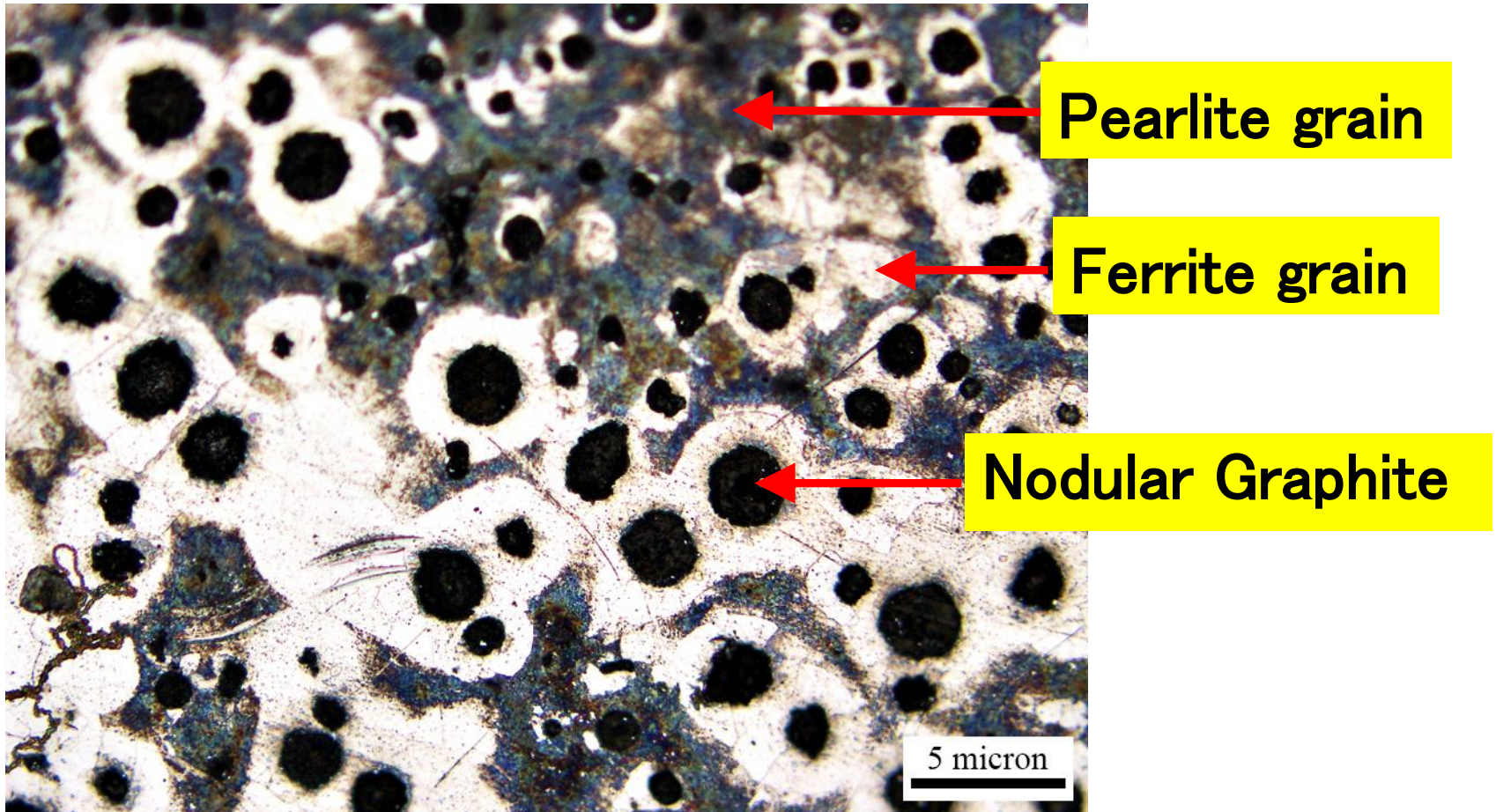
Rosette graphite

Ferrite grain

Ductile Cast Iron

- ❑ Ductile cast iron with consist of **carbon** content about **4.2–4.6wt.%** which eutectic composition
- ❑ Microstructure consist of ferrite, pearlite and graphite, which **graphite nodular shape**
- ❑ High strength and modulus of elasticity

Ductile Cast Iron



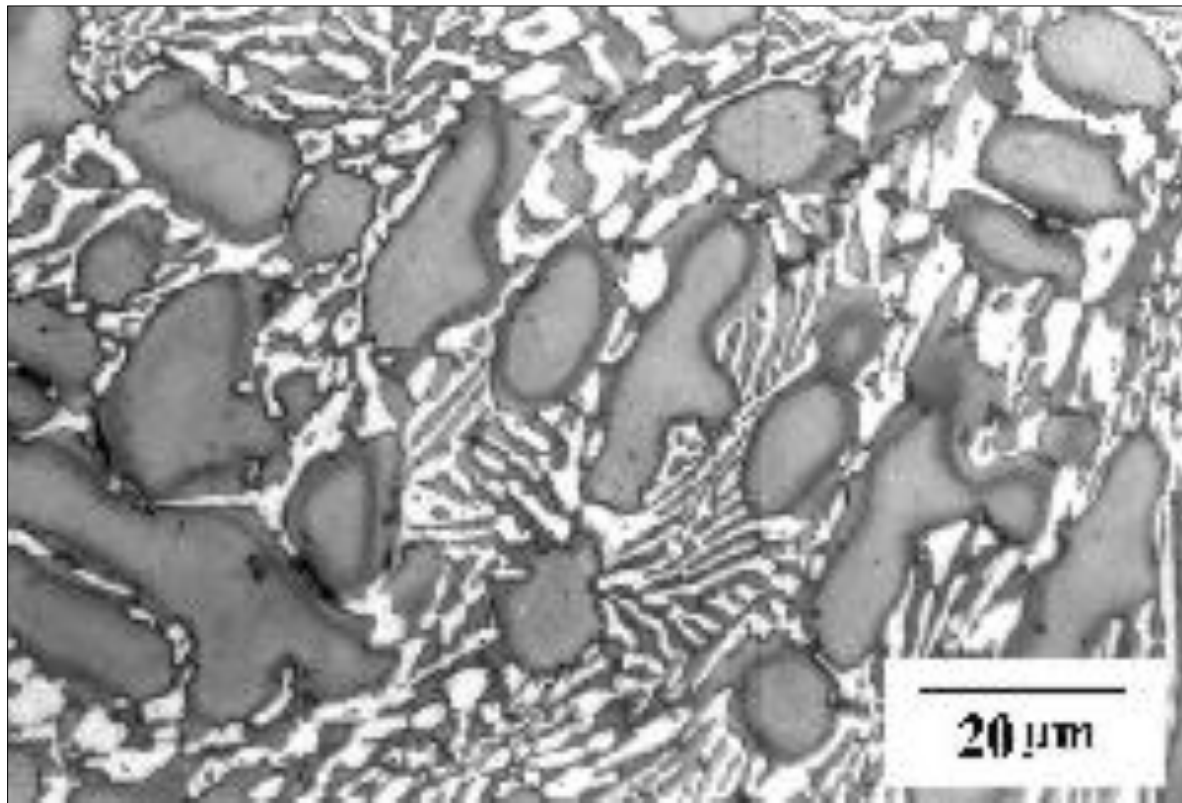
Alloys or Special Cast Iron

- ❑ Cast iron added some element such as **Cr** , **Si** , **Ni**, **Cu** since one or more element for improving physical and mechanical properties
- ❑ **Ni-resist** using for thermal and corrosion resistance

Alloys or Special Cast Iron

Wear resistant Hi-Cr cast iron structure
(Fe-2.6C-17Cr-2Mo-2Ni wt%)

Chromium Carbide (M_7C_3) is white



Comparing mechanical properties by ASTM Standard

Cast iron	Microstructure	UTS, ksi	YS, ksi	%Elongation
Gray	Ferrite +Pearlite	18	—	—
	Pearlite	30	—	—
	Fine Pearlite	40	—	—
Ductile	Ferrite +Pearlite	80	55	6
Malleable	Ferrite + Graphite	54	35	18
	Pearlite	105	90	10
High alloy	Ductile iron	55	30	20

Comparing mechanical properties by ASTM Standard

Cast iron	Process	Graphite	Matrix	Ductility
White	Rapid cooling rates	Carbide	Ferrite	No
Gray	Slow cooling rates	Flake	Ferrite/Pearlite	No
Malleable	White+annealing	Flake to nodule	Ferrite/Pearlite	Good
Nodular	Mg +slow cooling	Nodular	Ferrite/Pearlite	Good
Alloy	alloying element	Carbide	Ferrite/Pearlite	Good

Nonferrous Metals

- ❑ **Non-ferrous metal** is a metal which is not ferrous, including alloys, that does not contain **iron** in appreciable amounts. Generally more expensive than ferrous metals
- ❑ **Non-ferrous metals** are used because of desirable properties such as low weight (**Aluminium**), higher conductivity (**Copper**)



Nonferrous Metals

□ **Heavy Metals** is metal at density higher 4 g/cm³

□ **Light Metals** is metal at density from 1.74 to 4.5 g/cm³



Nonferrous Metals

- Only a small number are used in quantity as **the basis for engineering materials**, there are

1. Copper

2. Nickel

3. Zinc

4. Aluminium

5. Magnesium

6. Titanium

Heavy metal

Light metal

- Many others are used as **alloying elements** or special purpose

Nonferrous Metals

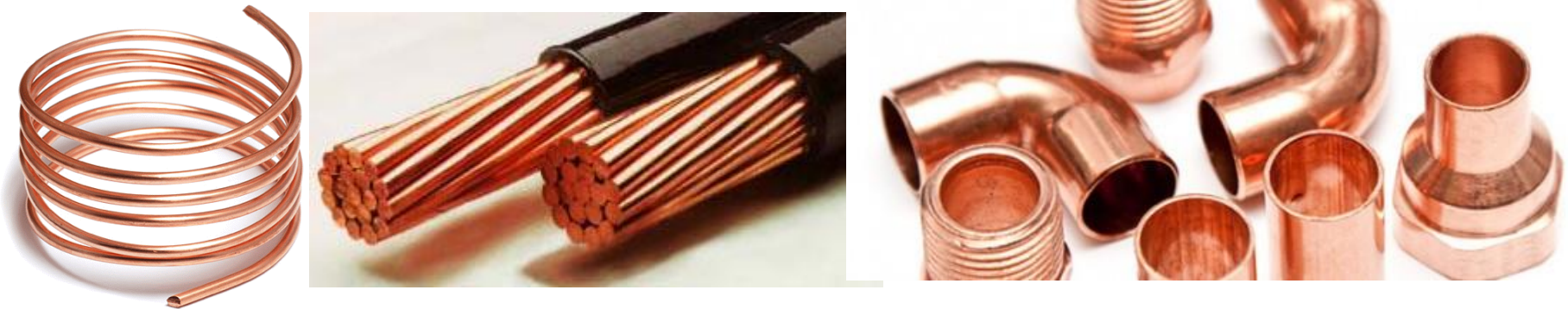
□ Copper (Cu)

- Density 8.9 g/cm³
- Melting point 1,073–1,093 °C
- Good thermal and electric conductivity
- Corrosion resistance

Nonferrous Metals

□ Copper (Cu)

- Copper sulphide (Cu_2S)
- Application: wire, cooling appliances, coil



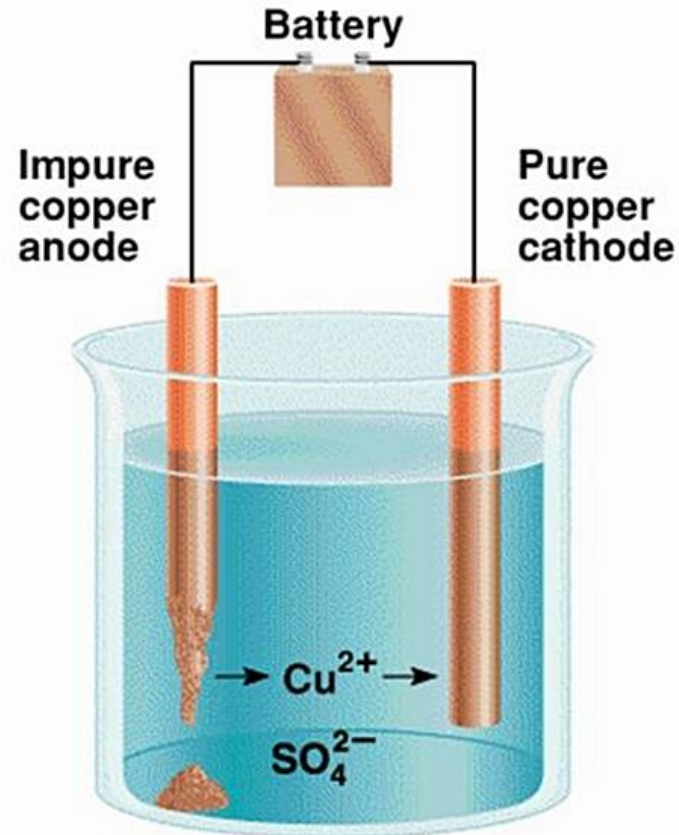
Source: www.copper.org

Nonferrous Metals

□ Copper (Cu)

Electrolytic Purification of Copper

High purity copper for using
in electric and electronic
"electrochemical method"



Nonferrous Metals

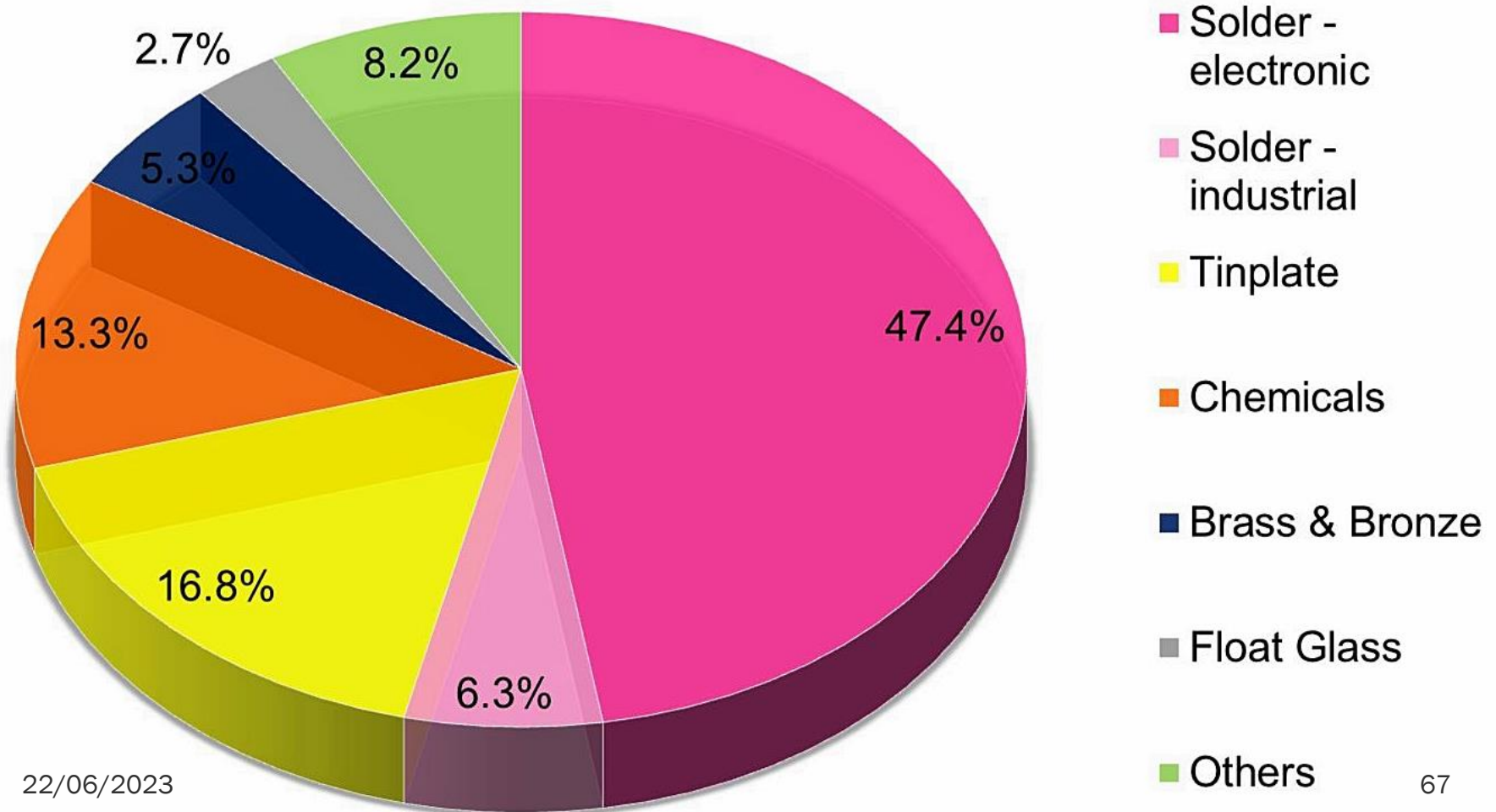
□ Tin (Sn)

- Soft and corrosion resistant
- Used to alloy with lead to make solder
- Used to alloy with copper to produce **bronzes**, e.g. **Bell metal** , **Candlestick**



Nonferrous Metals

World refined tin use by application, 2009



Nonferrous Metals

□ Lead (Pb)

- High Density
- Corrosion Resistant
- Malleable, but not ductile
- Very soft and weak

Lead applications



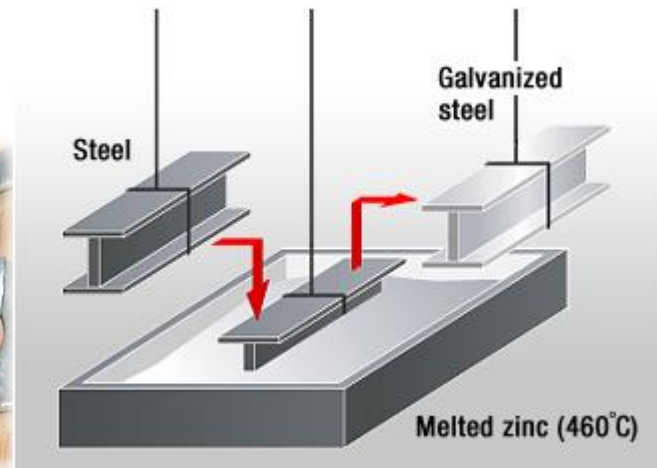
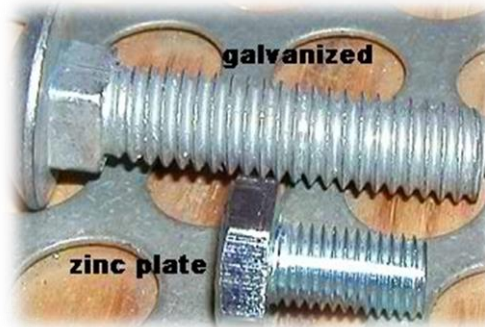
Nonferrous Metals

□ Zinc (Zn)

- No structural engineering uses in pure state
- Used as sacrificial anodes to protect steel
- Used as an alloy with copper to produce **brass**
- Used to corrosion coat steel by:

- Galvanising or

- Plating



Nonferrous Metals

□ Brass

- made of copper (Cu) and zinc (Zn)
- higher malleability than bronze or zinc
- low melting point of brass (900 to 940 °C)
- density is approximately 8.4 – 8.73 g/cm³
- normal brass (Zn ≤ 35%)
- Zn ≤ 20% is call red brass (can cold working)
- Zn 20–35% is call yellow brass (corrosion resistance lower than red brass)



Nonferrous Metals

□ Magnesium (Mg)

- Light weight
- Burns easily when being machined
- Used with Aluminium to make lightweight alloys

Magnesium Wheel



Nonferrous Metals

□ Lithium (Li)

- Very light
- Used to make alloys with aluminium for aerospace applications
- Used in batteries for mobile phones etc.



Nonferrous Metals

□ Titanium (Ti)

- Low density
- Similar strengths to steels
- Highly corrosion resistant
- Melts at 1670 °C
- Use restricted to < 400 °C

Used in

- Aerospace parts
- Racing cars, motorcycles and bicycles
- Pipework and heat exchangers in chemical plants and oil installations.

Titanium heat-exchangers



Nonferrous Metals

□ Nickel (Ni)

- Corrosion resistant – forms strong oxide layer
- Brittle in the pure state
- Used in stainless steels
- Alloyed with **copper** to produce **cupronickels** for coinage and **Monel** for corrosion-resistant valve parts
- Used in **Nimonic** and superalloys for aerospace engine parts



cupronickels



Monel



Nimonic

Nonferrous Metals

□ Chromium (Cr)

- Corrosion resistant – forms strong oxide layer
- Brittle in the pure state
- Used in stainless steels
- Used with Ni & Mo to produce strong steel alloys for bicycle and motorcycle frame tubes

